

UAV State Estimation and Motion Modeling System

User Documentation (documentation.pdf)

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Purpose. This document is the user manual for the UAV EKF State Estimation project. It explains exactly how to install prerequisites, run the code, choose datasets, and interpret outputs. Per the ENAE380 Final Project Guidelines, this document is designed to be sufficient for the instructor/TA to run the project without needing to contact the author.

1 What This Project Does

This project implements an Extended Kalman Filter (EKF)-based state estimation pipeline for UAV motion using CSV logs. The system supports:

- **Baseline mode (no heading):** estimates planar position and velocity using measurements $\mathbf{z} = [x, y]^T$ and state $\mathbf{x} = [x, y, v_x, v_y]^T$.
- **Heading-extended mode:** if the dataset includes yaw/yaw-rate fields, the driver can run an extended estimator for 4 or 6 degrees of freedom(DoF) using $\mathbf{z} = [x, y, \psi, \omega]^T$ and $\mathbf{x} = [x, y, v_x, v_y, \psi, \omega]^T$.
- **Multiple datasets:** works with simulated logs and “AERPAW-style” logs.
- **Plots + metrics:** generates trajectory plots and (if truth exists) error metrics such as root mean square error(RMSE).

2 Files Included (What You Need)

Place the following files in one folder (same directory) before running:

- `EKF.py` — estimator library module
- `run_uav_EKF.py` — main driver script (CSV-based testing and plotting)

- `app.py` (optional) — UI-based runner (if used) to have a more user friendly and interactive experience
- Sample CSV datasets (examples): `trajectory_log.csv`, `expo1.csv`
- Notebooks used during development (optional): `EKF_test.ipynb`, `test_code.ipynb` (used for data collection and log keeping)

3 Software Requirements (Install Everything)

This project was developed for Python 3 and uses common scientific Python packages.

3.1 Install Python 3

If Python is not installed:

- Windows/macOS: install Python from **python.org** (<https://www.python.org/downloads/>)
- Verify installation:

```
python --version
# or on some systems:
python3 --version
```

3.2 Create a Virtual Environment (Recommended)

Using a virtual environment prevents package conflicts:

```
# Create venv
python -m venv venv

# Activate (Windows)
venv\Scripts\activate

# Activate (macOS/Linux)
source venv/bin/activate
```

3.3 Install Required Libraries

Required libraries:

- `numpy` — array/matrix math

- `pandas` — CSV reading and processing
- `matplotlib` — plotting

Install:

```
pip install numpy pandas matplotlib
```

3.4 Library Resources (Official Docs)

If you need references for usage or troubleshooting:

- NumPy documentation: <https://numpy.org/doc/>
- Pandas documentation: <https://pandas.pydata.org/docs/>
- Matplotlib documentation: <https://matplotlib.org/stable/contents.html>

3.5 Optional Library (If Using the UI)

If you use the optional `app.py` interface via Streamlit, install:

```
pip install streamlit
```

Streamlit docs: <https://docs.streamlit.io/>

4 CSV Input Format (Required vs Optional Columns)

The driver script operates on CSV files with time-stamped measurements.

4.1 Required Columns

Every CSV must include:

- `t` — time (seconds)
- `x_meas`, `y_meas` — measured position

4.2 Optional Columns (If Available)

- `x_true`, `y_true` — ground truth (simulation validation)
- `psi_meas`, `omega_meas` — yaw (rad) and yaw rate (rad/s)

Automatic mode selection: `run_uav_EKF.py` checks whether `psi_meas` and `omega_meas` exist. If they do, it runs the 6-state estimator; otherwise, it runs the 4-state estimator.

5 How to Run (Step-by-Step)

5.1 Step 1: Put Files in One Folder

Example folder contents:

- `EKF.py`
- `run_uav_EKF.py`
- `trajectory_log.csv` (or another dataset)

5.2 Step 2: Choose Which CSV to Run

Open `run_uav_EKF.py` and set the dataset path in the script (near the top). Example:

```
CSV_FILE = "trajectory_log.csv"
# or:
CSV_FILE = "expo1.csv"
```

5.3 Step 3: Execute

Run from the same directory:

```
python run_uav_EKF.py
```

5.4 What You Should See

On success, the script prints the detected columns and whether heading fields were found, then produces plots:

- Measured vs EKF position track
- Truth vs EKF track (if truth exists)
- Velocity estimates (optional)
- Innovation/diagnostics (optional)

6 Common Issues and Fixes

6.1 “ModuleNotFoundError” for numpy/pandas/matplotlib

Fix by installing packages in the active environment:

```
pip install numpy pandas matplotlib
```

If using a venv, ensure it is activated before installing and running.

6.2 File Not Found / Wrong Working Directory

Make sure your terminal is in the same folder as `run_uav_EKF.py` and the CSV:

```
ls
# or on Windows:
dir
```

6.3 Plots Look “Unchanged” (EKF Seems to Do Nothing)

This usually means the filter is tuned to heavily trust measurements. Adjust:

- Increase process noise parameter (q) to allow more maneuvering responsiveness.
- Increase measurement noise (R) to trust measurements less.

The design report explains the tuning process and includes recommended values.

6.4 Irregular dt / Discontinuities in Real Data

Real logs may contain repeated timestamps or jumps. The driver script clamps Δt to remain positive. If results are still unstable, use a dataset segment with fewer dropouts, or apply innovation gating (described in the design report).

7 Video Walkthrough

- project file structure
- running the driver script on multiple datasets
- explanation of estimator outputs and plots
- Can be found here: https://drive.google.com/file/d/1ebdRJMG_J40cQ1fqVbgYtBxwZNLoF4Jd/view?usp=sharing